

APO-METOPROLOL

Metoprolol Tartrate Tablets USP 100mg

Anti-anginal and Antihypertensive Agent

Pharmacology: APO-METOPROLOL (metoprolol tartrate) is a beta-adrenergic blocking agent with predominant blocking effect on beta₁-receptors.

Although the mechanism of the antihypertensive effect has not been established with certainty, the following factors may be involved:

- a) competitive ability to antagonize catecholamine-induced tachycardia at the beta-receptor sites in the heart, thus decreasing cardiac output,
- b) inhibition of renin release by the kidney,
- c) inhibition of the vasomotor centres.

The mechanism of the anti-anginal effect is also uncertain. An important factor may be the reduction of myocardial oxygen requirements by blocking catecholamine-induced increases in heart rate, systolic pressure, and the velocity and extent of myocardial contraction.

In man, metoprolol is rapidly and nearly completely absorbed from the gastrointestinal tract. Due to the large hepatic first pass elimination, only half of the dose reaches the systemic circulation. This fraction increases to 70-80% with prolonged therapy. Metoprolol is extensively distributed to extravascular tissues. The maximum plasma concentrations are reached within 1½ - 2 hours. Binding to serum protein is about 10%. The plasma half-life is approximately 3 hours. The elimination rate is not dose-dependent in the therapeutic range.

In humans, metoprolol is metabolized into three main urinary metabolites. Two of these have some pharmacological activity which do not contribute significantly to the beta-blocking effect of the parent compound.

More than 90% of an oral therapeutic dose is recovered in the urine within 72 hours. Metoprolol is excreted mainly by glomerular filtration.

Indications: Hypertension: In the treatment of patients with mild or moderate hypertension. It is usually used in combination with other drugs, particularly a thiazide diuretic. However, it may be tried alone as an initial agent in those patients in whom, in the judgment of the physician, treatment should be started with a beta-blocker rather than a diuretic.

The combination of metoprolol with a diuretic or peripheral vasodilator has been found to be compatible and generally more effective than metoprolol alone. Experience with most commonly used antihypertensive agents has not given evidence of incompatibility.

Metoprolol is not recommended for the emergency treatment of hypertensive crises.

Angina Pectoris: For the prophylaxis of angina-pectoris due to ischemic heart disease.

Contraindications: Sinus bradycardia, second and third degree AV block, right ventricular failure secondary to pulmonary hypertension, congestive heart failure, cardiogenic shock, anesthesia with agents that produce myocardial depression, e.g., ether.

Warnings: Cardiac Failure: Special caution should be exercised when administering metoprolol to patients with a history of heart failure. Sympathetic stimulation is a vital component supporting circulatory function in congestive heart failure, and inhibition by means of beta-adrenergic blockade carries the potential hazard of further depressing myocardial contractility and precipitating cardiac failure.

Abrupt Cessation of Therapy: Patients with angina pectoris should be warned against abrupt discontinuance of metoprolol. There have been reports of severe exacerbation of angina and the occurrence of myocardial infarction or ventricular arrhythmias, and of myocardial infarction or ventricular arrhythmias occurring in some patients with angina pectoris, following abrupt discontinuance of beta-blocker

therapy. The last two complications may occur with or without preceding exacerbation of angina pectoris. Therefore, when discontinuance of metoprolol is planned in patients with angina pectoris, the dosage should be gradually reduced over a period of about 2 weeks and the patient should be carefully observed. The same frequency of administration should be maintained. In situations of greater urgency, metoprolol dosage should be discontinued stepwise and under conditions of closer observation. If angina markedly worsens or acute coronary insufficiency develops, it is recommended that treatment with metoprolol be reinstituted promptly, at least temporarily.

Oculomucocutaneous Syndrome: Various skin rashes and conjunctival xerosis have been reported in treatment with beta-blockers, including metoprolol. A severe oculomucocutaneous syndrome, whose signs include conjunctivitis sicca and psoria-siform rashes, otitis, and sclerosing serositis has occurred with the chronic use of one beta-adrenergic blocking agent (proctolol). This syndrome has not been observed with metoprolol or any other such agent. However, physicians should be alert to the possibility of such reactions and discontinue treatment if they occur.

Bradycardia: Severe sinus bradycardia may occur with the use of metoprolol due to unopposed vagal activity remaining after blockade of beta-adrenergic receptors. In such cases, dosage should be reduced.

In Patients with Thyrotoxicosis: Possible deleterious effects from long-term use of metoprolol have not been adequately appraised. Beta-blockade may the clinical signs of continuing hyperthyroidism or its complications, and give a false impression of improvement. Therefore, abrupt withdrawal of metoprolol may be followed by an exacerbation of the symptoms of hyperthyroidism, including thyroid storm.

Precautions: Because of its relatively greater beta₁-blocking effect, metoprolol may be tried in patients with diseases associated with bronchospasm who require beta-blocker therapy. Careful monitoring of such patients is mandatory. Since beta₁ selectivity is not absolute, a sympathomimetic bronchodilator should be used concomitantly. In patients already on bronchodilator therapy, the dose may have to be increased. In spite of these precautions, the patients' respiratory status may worsen, and in such cases metoprolol should be withdrawn. In Patients Undergoing Elective or Emergency Surgery: The management of patients being treated with beta-blockers and undergoing elective or emergency surgery is controversial. Although beta-adrenergic receptor blockade impairs the ability of the heart to respond to beta-adrenergically mediated reflex stimuli, abrupt discontinuance of the therapy with metoprolol may be followed by severe complications (see warnings). Some patients receiving beta-adrenergic blocking agents have been subject to protracted severe hypotension during anesthesia. Difficult in restarting and maintaining the heartbeat has also been reported. For these reasons, in patients with angina undergoing elective surgery metoprolol should be withdrawn gradually following the recommendation given under Abrupt Cessation of Therapy (see warnings).

All clinical and physiological effects of beta-blockade are no longer present 48 hours after cessation of medication. In cases of emergency surgery, since metoprolol is a competitive inhibitor of beta-adrenergic receptor agonist, its effects may be reversed, if necessary by sufficient doses of such agonists as isoproterenol or levaterenol. Suitable laboratory test should be carried out at appropriate intervals. Caution should be observed particularly in patients with impaired hepatic function. Patients with signs of severely compromised liver function should be treated with reduced dosage of metoprolol. Pregnancy: Since metoprolol has not been studied in human pregnancy, the drug should not be given to pregnant women. The use of any drug in patients of child bearing potential requires that the anticipated benefit be weighed against possible hazards. Children: There is no experience in the treatment of pediatric age groups.

Drug Interactions: Metoprolol acts selectively without abolishing the inotropic action of digitalis on the heart muscle. However, the positive inotropic action of digitalis may be reduced by the negative inotropic

effect of metoprolol when the 2 drugs are used concomitantly. The effects of beta-blockers and digitalis are additive in depressing AV conduction. In patients without a history of cardiac failure continued depression of the myocardium over a period of time can, in some cases, lead to cardiac failure. Therefore, at the first sign or symptom of impending cardiac failure, patients should be fully digitalized and /or given a diuretic, and the response observed closely. If cardiac failure continues despite adequate digitalization and diuretic therapy, metoprolol should be withdrawn immediately.

Metoprolol should be administered with caution to patients subject to spontaneous hypoglycemia, or to diabetic patients (especially those with labile diabetes) who are receiving insulin or oral hypoglycemic agents. Beta-adrenergic blockers may mask the premonitory signs and symptoms of acute hyperglycemia. The combination of metoprolol with other antihypertensive agents produces a greater fall in blood pressure than either drug alone. This occurs regardless of which drug is administered first. The same degree of blood pressure control may be achieved by lower than usual doses of each drug. Therefore, when using combination therapy careful monitoring of the dosages is required until the blood pressure of the patient is stabilized (see dosage). Patients receiving catecholamine depleting drugs such as reserpine or guanethidine, should be closely monitored if metoprolol is administered as the added beta-adrenergic blocking action may produce an excessive reduction of sympathetic activity. Metoprolol should not be given with other beta-blockers.

Concurrent administration of cimetidine and metoprolol was reported to increase significantly the plasma levels of metoprolol. **Epinephrine and beta-blockers:** There may be increased difficulty in treating an allergic type reaction in patients on beta-blockers. In these patients, the reaction may be more severe due to pharmacologic effects of the beta-blockers and problems with fluid changes. Epinephrine should be administered with caution since it may not have its usual effects in the treatment of anaphylaxis. On the one hand, larger doses of epinephrine may be needed to overcome the bronchospasm, while on the other hand, these doses can be associated with excessive alpha adrenergic stimulation with consequent hypertension, reflex bradycardia and heart block and possible potentiation of bronchospasm. Alternatives to the use of large doses of epinephrine include vigorous supportive care such as fluids and the use of beta agonists including parenteral salbutamol or isoproterenol to overcome bronchospasm and norepinephrine to overcome hypotension.

Adverse Effects: The most common adverse reactions reported were tiredness, dizziness and gastrointestinal symptoms. The most severe adverse reactions reported were congestive heart failure and bronchospasm. **Cardiovascular:** congestive heart failure (see warnings), severe bradycardia, lengthening of PR interval, second and third degree AV block, sinus arrest, palpitations, chest pains, cold extremities. Raynaud's phenomenon, claudication peripheral vasospasm, hot flushes.

CNS: headache, dizziness, insomnia, mental depression, lightheadedness, anxiety, tinnitus, weakness, sedation, vivid dreams with or without insomnia, vertigo, paresthesia, lethargy, drowsiness, fatigue, anorexia, poor concentration. **Gastrointestinal:** diarrhea, constipation, flatulence, heartburn, nausea and vomiting, abdominal pain, dryness of mouth. **Respiratory:** shortness of breath, wheezing, bronchospasm, status asthmaticus. **Allergic/Dermatological (see Warning):** exanthema, sweating, pruritus, psoriasiform rash, urticaria, dry skin.

Ophthalmological: blurred vision and non-specific visual disturbances, itching eyes, dry eyes, conjunctivitis. **Miscellaneous:** tiredness, weight gain, decrease in libido (very rarely), hearing loss. **Clinical Laboratory Test Findings:** elevated (rarely) transaminases, BUN, alkaline phosphatase and lactic dehydrogenase. Thrombocytopenia and leukopenia have been reported rarely.

Symptoms and Treatment of Overdosage: The most common signs to be expected with overdosage of a beta-adrenergic blocking agent are bradycardia, congestive heart failure, hypotension, bronchospasm,

and hypoglycemia. If overdosage occurs, therapy with Apo-Metoprolol should be discontinued, and the patient observed closely. In addition, if required, the following therapeutic measures are suggested : **Bradycardia:** Atropine or another anticholinergic drug.

Heart block (second degree or total): Isoproterenol or transvenous cardiac pacemaker.

Congestive heart failure: conventional therapy.

Hypotension (depending on associated factors): epinephrine rather than isoproterenol or norepinephrine may be useful in addition to atropine and digitalis.

Bronchospasm: administer isoproterenol and aminophylline.

Hypoglycemia: intravenous glucose.

It should be remembered that metoprolol is a competitive antagonist of isoproterenol and hence large doses of isoproterenol can be expected to reverse many of the effects of excessive doses of metoprolol. However, the complications of excess isoproterenol should not be overlooked.

Dosage: **Hypertension:** Metoprolol should be administered as 2 equal doses per day, morning and evening. Metoprolol is usually used in conjunction with other antihypertensive agents, particularly a thiazide diuretic, but may be used alone (see Indications). The dose must always be adjusted to the individual requirements of the patients, in accordance with the following guidelines. Treatment should be initiated with doses of 50mg twice daily. A small number of patients may respond satisfactorily to this dosage. However, if an adequate response is not obtained after 1 week, dosage should be increased to 100mg twice daily. In some cases the daily dosage may have to be increased by further 100mg increments of not less than 2 weeks, up to a maximum of 200mg twice daily, which should not be exceeded. The usual maintenance dose is in the range of 150 to 300mg daily. When metoprolol is combined with another antihypertensive agent which is already being administered, it should be added initially at a dose of 50mg twice daily. After one or two weeks the daily dose may be increased, if required, in increments of 100mg at intervals of not less than two weeks until adequate blood pressure control is obtained.

Angina Pectoris: 100 to 400mg/day in divided doses. Treatment should be initiated with 50mg twice daily for the first week. If the response is not adequate, the daily dosage should be increased by 100mg for the next week. The usual maintenance dose is 200mg/day. The need for further increase in dosage should be closely monitored at weekly intervals, and the dosage increased in 100mg daily increments to a maximum of 400mg/day in 2 or 3 divided doses. A metoprolol dose of 400mg/day should not be exceeded.

Supplied: Each round, white, biconvex tablets, scored and engraved "APO" over "M100" on one side, plain on the other contains metoprolol tartrate 100mg. Available in PVC blisters of 30's and 500's tablets.

Storage Conditions: Store below 30°C.

Manufacturer: Apotex Inc., 150 Signet Drive, Toronto Ontario, Canada M9L 1T9.

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